

3. Develop a Secure User Login and Session Management System using Java Servlets, Cookies, and HttpSession for user authentication and access control.

Index.html

```
<!DOCTYPE html>

<html>

  <head>

    <title>Login Page</title>

  </head>

  <body>

    <h1>User Login</h1>

    <form action="login" method="post">

      Username: <input type="text" name="username"><br>

      Password: <input type="password" name="pass"><br>

      <button type="submit">Login</button>

    </form>

  </body>

</html>
```

LoginServlet.java

```
package com.og.practical3c;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
import java.io.IOException;
import java.sql.Connection;
```

```

import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

@WebServlet("/login")

public class LoginServlet extends HttpServlet {

    @Override

    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws
ServletException, IOException {

        String username = req.getParameter("username");

        String password = req.getParameter("pass");

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/jdbc_demo","root","root");

            PreparedStatement ps = con.prepareStatement("select * from users where username=?
and password=?");

            ps.setString(1, username);

            ps.setString(2, password);

            ResultSet rs = ps.executeQuery();

            if(rs.next()){

                HttpSession session = req.getSession();

                session.setAttribute("user", username);

                resp.sendRedirect("home");

            }else{

                resp.getWriter().println("<h1>Incorrect Username and Password</h1>");

            }

        }

```

```

        } catch (Exception ex) {
            e.printStackTrace();
        }
    }
}

```

HomeServlet.java

```
package com.og.practical3c;
```

```

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
import java.io.IOException;

```

```
@WebServlet("/home")
```

```
public class HomeServlet extends HttpServlet{
```

```
    @Override
```

```
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws
ServletException, IOException {
```

```
        HttpSession session = req.getSession(false);
```

```
        String username = session.getAttribute("user").toString();
```

```
        resp.getWriter().println("<h1> Welcome "+username+" Login Successful.</h1>");
```

```
    }
```

```
}
```